The Utah Teapot in CONV Geometry: Nodes on the Surface, Rays by Transport

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Abstract—We rebuild the Utah Teapot from points that lie on its own surface. The representation is a lattice of nodes reconstructed by Compact Ordered Nodal Variation (CONV) with ledger-free admission. Three additions make it a geometry and not only an image filter. Declared seams with cubic-exact one-sided stencils let the lattice reproduce bicubic Bézier patches exactly: with four node intervals per patch, 560 on-surface nodes reproduce the 28 original patches to 2×10^{-15}. A symmetric bubble limiter replaces the standard one, which is discontinuous in the data and can open cracks in a surface; the new limiter is continuous and keeps the variation bound. And because a CONV cell depends on a fixed block of nodes, the surface has a pointwise evaluator with exact derivatives. We render it by correspondence transport, which sends a lattice of known parameters forward to the screen and lets each pixel correct a prediction with a Newton step; it is 1.6 to 2.0 times faster than a per-ray tracer of the same surface and agrees with a fine reference on 99.9% of surface pixels. The representation needs no patches, control points, degrees or knots, only points on a surface, so later versions enter the same way: the 2026 Utah Teapot is rebuilt from 9 by 9 samples of each of its surfaces, interior included. We then measure against the defects that the later teapots were made to fix. A node lattice makes the body 233 times rounder with the same node count, renders without a mesh, and carries the 2026 shape with normals matched to 0.11° on average across joints. It does not reproduce that model's curvature continuity, its trimming or its chamfers. We report those as they are.

Index Terms—Utah Teapot, interpolating surface, variation diminishing, Bézier patch, ray tracing parametric surfaces, correspondence transport.

I. Introduction

The Utah Teapot was modelled by Martin Newell in 1974–75 as 28 bicubic Bézier patches and scaled to its familiar proportions by Jim Blinn in 1976 [1], [2], [3], [4]. Its control points mostly lie off the surface: a Bézier patch passes through only four of its sixteen. Every later version of the model addressed something that the original patches leave open. Crow added a bottom. Driskill made the body round, trimmed the handle and spout to the body and added an interior. Fish improved the interior. Yuksel's 2026 version adds chamfers, a lighter interior, texture layouts and curvature continuity for smooth reflections [5], [6].

This paper asks what the teapot looks like in a different kind of representation: one whose every control value is a point of the surface. CONV [8] interpolates samples on a regular lattice by a piecewise quintic whose derivative cannot change sign more often than the samples do. It was built for images. Used for positions, it gives a surface through a lattice of nodes, and a node can be moved, measured, textured or collided with as the surface point it is.

The contributions are these.

1) Seamed CONV lattices (Section III). A seam cuts the difference stencils along a node row or column. With cubic-exact end stencils, a seamed lattice reproduces any piecewise bicubic surface whose patch joints lie on seams, wherever admission leaves the proposal alone. For the teapot it is alone everywhere.
2) A continuous bubble limiter (Section IV). We show that the standard limiter jumps, give a symmetric replacement, and prove that it keeps the variation bound.
3) A pointwise surface evaluator and transport ray tracing (Sections V and VIII), with a measured comparison against per-ray tracing of the same surface.
4) An account against the later teapots (Section VII): what the representation improves, what it leaves unchanged, and what it does not address.

An engine that implements all of it accompanies the paper and runs in a browser on one CPU thread. Its controls follow the official Utah Teapot page: a choice of version (1975, 1976, 1987, 2026), switches for the handle, spout, lid, body and interior, Blinn's scale, wireframe, texture coordinates per part or per patch, and reflection lines; it adds gloss and a rigid-body drop.

A. For readers of the spline literature

CONV on a lattice is an interpolating surface formulation, and the questions usually put to one have short answers here.

- Interpolation. The surface passes through every node. There are no off-surface controls and no handles.
- Locality and cost. A cell depends on the six samples around it in each direction (eight next to a seam). Every step is closed form, with constant work per cell and no global solve.
- Precision. Linear data are reproduced exactly, so straight pieces need no special constraint. With seams, cubics are reproduced exactly (Proposition 1).
- Continuity. C^2 across node lines wherever the limiter is inactive; C^0 across a seam, which is how a crease is modelled. Curvature is not controlled at a seam, and Section VII measures what that costs.
- Shape. The derivative along a line changes sign no more often than the samples do. Overshoot next to a

crease or a step is excluded by construction, not by tuning.
- Dependence on the data. Continuous and piecewise affine in the nodes (Theorem 2).
- What it lacks. Samples are taken as uniformly spaced in the parameter. Coordinates are limited separately, so where the limiter acts the surface is not rotation invariant. It is not a subdivision or a convex-hull scheme.

II. CONV Lines

We restate what is needed from [8]. For samples y_i on a line, with secants $\delta_i = y_{i+1} - y_i$, each node receives a scaled jet (m_i, q_i) from a difference stencil. The cell $[i, i+1]$ gets five raw currents, the differences of six quintic Bézier controls:

$$\begin{aligned} a_{i0} &= \tfrac{m_i}{5}, & a_{i1} &= \tfrac{m_i}{5} + \tfrac{q_i}{20}, \\ a_{i4} &= \tfrac{m_{i+1}}{5}, & a_{i3} &= \tfrac{m_{i+1}}{5} - \tfrac{q_{i+1}}{20}, \end{aligned} \tag{1}$$

and $a_{i2} = \delta_i - a_{i0} - a_{i1} - a_{i3} - a_{i4}$. Ledger-free admission clips a_i into a box built from the two neighbouring secants and terminal currents, and restores the cell mass δ_i by one capped common shift. The admitted currents c_i satisfy $\mathrm{var}(c^\flat) \le \mathrm{var}(\delta)$ for any proposal whose cells sum to their secants; the proof in [8] uses nothing else. The cell is then

$$y(i+t) = y_i + \sum_{j=0}^{4} c_{ij}\, \tau_j(t) + \omega_i\, \psi(t), \tag{2}$$

with $\tau_j = \sum_{r>j} B_r^5$ and the odd bubble $\psi(t) = [t(1-t)]^3(2t-1)$. In the interior of a line the jets are the fourth-order centred stencils, and a cell depends on the six samples $y_{i-2}, \dots, y_{i+3}$ only.

III. Seams and Cubic Precision

A first attempt to put the teapot on a plain CONV lattice fails near its creases and patch joints. With four node intervals per patch the worst deviation from the Bézier surface is 5.9×10^{-2} on a body 4.0 wide, against 1.05×10^{-1} for the same nodes joined bilinearly; next to the crease of a flat bottom cap, CONV is worse than linear. The variation bound holds, and there is no overshoot. The loss is in the jets: a centred stencil that straddles a crease measures a slope that belongs to neither side.

A. Seams

A seam is a node row or column across which no difference stencil may reach. Values remain continuous across it. The two ends of an open line are seams; a closed ring without seams is periodic. A node on a seam has two jets, one from each side. For a segment of at least four nodes between seams we use the stencils that are exact for cubics and still fit the six-sample window:

$$\begin{aligned} m_0 &= \tfrac{1}{6}(-11y_0 + 18y_1 - 9y_2 + 2y_3), \\ q_0 &= 2y_0 - 5y_1 + 4y_2 - y_3, \\ m_1 &= \tfrac{1}{6}(-2y_0 - 3y_1 + 6y_2 - y_3), \\ q_1 &= y_0 - 2y_1 + y_2, \end{aligned} \tag{3}$$

with their mirror images at the other end and the centred stencils from the third node on. Admission is unchanged, except that the terminal current a cell sees in its neighbour across a seam, $a_{i-1,4}$ or $a_{i+1,0}$, is now the neighbour's own one-sided value. This is the case for which the knot-exclusivity terms of the box were written; without seams they collapse to the sign of a shared slope. A cell next to a seam depends on $y_{i-3}, \dots, y_{i+4}$.

Proposition 1 (Cubic precision). Let a segment between two seams have at least four nodes, and let its samples be those of a cubic polynomial at equally spaced parameters. If admission leaves every raw current of the segment unchanged, the CONV line equals that cubic on the segment, with zero bubble.

Proof. The stencils (3) and the centred stencils return the first and second derivatives of a cubic exactly. The raw quintic of each cell is then the quintic Hermite interpolant of exact two-jets of a cubic at both ends, which is that cubic. With $c_i = a_i$ the second term of the bubble fit vanishes, and its first term is a fifth difference, which vanishes on a cubic. □

Theorem 1 (Exact Bézier reproduction). Let a surface be a grid of bicubic Bézier patches, sampled at $n \ge 3$ equal parameter intervals per patch in each direction, with seams on every patch boundary. If admission is inactive in both factors, the CONV surface of Section V equals the Bézier surface.

Proof. Along a node row inside one patch the samples are those of a cubic, so by Proposition 1 every row line equals the surface along that row at every u. The second factor then interpolates, at fixed u, values that are samples of the cubic $v \mapsto S(u, v)$ within the patch. □

The hypothesis is checkable, and for the teapot it holds. Table I gives the measured deviation at eight samples per node interval over all four parts. From four intervals per patch on, the teapot's own geometry passes admission untouched and the two surfaces agree to rounding. At three intervals a few cells are clipped. At two there are too few nodes for a cubic stencil.

TABLE I
Deviation from the Bézier teapot (body 4.0 wide)

		Seamed, cubic ends		Seamless	
n	Nodes	max	rms	max	rms
2	152	2.3×10^{-1}	5.4×10^{-2}	2.1×10^{-1}	4.8×10^{-2}
3	324	2.9×10^{-5}	1.3×10^{-6}	8.8×10^{-2}	1.7×10^{-2}
4	560	2.3×10^{-15}	4.1×10^{-16}	5.9×10^{-2}	9.5×10^{-3}
6	1224	3.4×10^{-15}	6.2×10^{-16}	3.5×10^{-2}	4.7×10^{-3}
8	2144	2.4×10^{-15}	4.1×10^{-16}	2.5×10^{-2}	2.9×10^{-3}

The node count is not smaller than the control-point count; 28 patches have 448 control points and the exact lattice has 560 nodes, 64 of them on a flat bottom cap that is not one of the 28 patches. What changes is where they are. All 560 are points of the teapot.

IV. A Continuous Bubble

The bubble amplitude ω_i is fitted and then limited so that the degree-six Bernstein coefficients of the cell's derivative keep their signs. Writing d_2, d_3, d_4 for the three coefficients the bubble touches, the enriched ones are $d_2 - \omega/5$, $d_3 + 2\omega/5$ and $d_4 - \omega/5$. The standard limiter [8] lets each keep its sign one-sidedly: for $d_2 > 0$ it requires only $\omega \le 5d_2$.

Proposition 2 (The standard limiter is discontinuous). Fix a fitted amplitude $\widehat{\omega} > 0$ and let d_2 pass through zero with d_3, d_4 fixed and nonzero. For $d_2 \downarrow 0$ the admitted amplitude tends to 0. For $d_2 \uparrow 0$ the constraint from d_2 is a lower bound that tends to 0, and the admitted amplitude tends to $\min\{\widehat{\omega}, U\}$, where $U > 0$ is the upper bound from the other two coefficients whenever one exists.

On a line this is a jump in the data-to-value map. On a surface it is a crack: the second factor's inputs vary continuously with u, and where one of its coefficients changes sign the surface position jumps. The defect was observed in an earlier cloth experiment; here is its cause and a cure.

Theorem 2 (Symmetric limiter). Let

$$M_i = \min\{5|d_{i2}|,\ \tfrac{5}{2}|d_{i3}|,\ 5|d_{i4}|\},$$
$$\omega_i = \max\{-M_i, \min\{M_i, \widehat{\omega}_i\}\}. \tag{4}$$

Then ω_i is a continuous piecewise-affine function of the cell's samples, no enriched coefficient changes sign, and the variation bound of [8] holds unchanged. Wherever $|\widehat{\omega}_i| \le M_i$ the amplitude equals the fitted one, so the leading-error reduction 1681/1950 is retained.

Proof. $|\omega_i| \le M_i$ gives $|\omega_i/5| \le |d_{i2}|$, $|2\omega_i/5| \le |d_{i3}|$ and $|\omega_i/5| \le |d_{i4}|$, so each enriched coefficient is zero or has the sign of the one it replaces. That is the only property of the amplitude used in the variation proof. $\widehat{\omega}_i$ is linear in the samples and the admitted currents, which are continuous and piecewise affine; M_i is a minimum of absolute values of linear forms of them; and a clamp is continuous in its three arguments. □

The symmetric bound is the tighter of the two, since it limits both signs of the amplitude. On resolved smooth data the fit is of fifth order in the spacing and the coefficients of first order, so neither limiter acts and the two agree. Table II measures the difference where it matters: the largest change of position between surface samples one step apart along u, on the teapot body with its nodes displaced by noise of standard deviation 0.08. A continuous surface gives a change proportional to the step. The standard limiter leaves a jump of 1.3×10^{-4} at a step of 10^{-6}, about a hundred times the continuous value.

TABLE II
Largest position change over one parameter step

Bubble	step 10^{-4}	step 10^{-6}
off	1.8×10^{-3}	1.4×10^{-6}
standard	2.0×10^{-3}	1.3×10^{-4}
symmetric	1.9×10^{-3}	1.4×10^{-6}

With the symmetric limiter the whole factor, bubble included, is continuous in the data, and the surface of the next section is watertight within each part.

V. The Surface

A part is a lattice of nodes $P_{kl} \in \mathbb{R}^3$, closed around (u) and open along (v). The surface is the fixed-order composition of two CONV factors, evaluated at a point. For (u, v) in cell (i, j) with local parameters (s, t):

1) for each node row $l = j - 3, \ldots, j + 4$ that exists, evaluate the row's CONV cell i at s, giving $R_l(s)$;
2) admit one CONV cell along v from the window $R_{j-3}, \ldots, R_{j+4}$ and evaluate it at t.

Each coordinate is treated separately in the object's own frame. The admitted currents of step 1 do not depend on s and are compiled once per lattice; step 2 is admitted at every query, because its inputs depend on u. A point costs 24 polynomial evaluations and three admissions. This is exactly the ordered composition of [8], not an approximation of it; the ledger form could not be evaluated this way, since its sign ledger needs the whole line.

Derivatives. Admission is piecewise affine, so forward-mode differentiation through its minima and maxima is exact wherever a derivative exists. We evaluate both steps on dual numbers that carry $\partial/\partial s$ and $\partial/\partial t$ and obtain S_u and S_v with the point. The normal is their cross product.

Regularity. Within a part the surface is continuous (Theorem 2). Where admission and the limiter are inactive it is C^2 across node lines away from seams, because adjacent cells share two-jets and ψ has triple zeros. Across a seam it is C^0 by design.

Invariance. Each factor commutes with translation and with scaling of a coordinate, so an axis-aligned affine map of the nodes maps the surface exactly; Blinn's vertical scaling of Newell's teapot is such a map. Where admission is inactive the operator is linear and commutes with every affine map. Where it is active, a rotation of the nodes does not in general rotate the surface, since coordinates are admitted separately.

VI. The Teapot

The source is the original data from the Utah Model Repository [7]: 28 patches, without a bottom. They assemble into four parts, each a grid of patches closed around: the body (rim, upper and lower body: four patches around, three along), the lid (knob and lid: four by two), the handle and the spout (two half tubes around, two along). We add an optional flat cap under the body, as Blinn suggested [4]; it meets the body at a 90° crease, which a seam keeps exactly, and the lowest point of the CONV body is the plane itself.

Three lattice styles are provided for this data.

- Exact: seams on every patch boundary, nodes at equal Bézier parameters. This is the 1976 teapot to rounding (Table I).
- Round: as exact along v, but every ring of the body and lid lies on a true circle and is closed without seams.
- Seamless: no seams, the closure of [8]. Smooth across the patch joints, at the deviation of Table I.

A. Other versions, from samples alone

Nothing in Section V asks where the nodes came from. The official page generates meshes of every version, and a mesh vertex of a patch is a point of its surface. We took, for the 2026 version, the 9×9 vertices of each of its 66 untrimmed surfaces at eight intervals per patch, and for the 1987 version the four patches of Crow's round bottom. Symmetry leaves 19 distinct patches of the 66. Each becomes one open lattice with cubic ends. No control point, degree, knot vector or continuity construction of the source model is used or known to us; the samples are not even uniform in the source parameter.

The versions offered are then: 1975, Newell's proportions, which is the 1976 lattice with its heights scaled by 4/3, an exact operation (Section V); 1976; 1987, the 1976 lattice raised by 0.15 with the sampled bottom; and 2026, 5346 nodes on 46 exterior and 20 interior surfaces. Adjacent lattices share their edge samples, and a CONV line along an edge depends on that edge's samples alone, so the lattices meet without gaps: the largest measured gap over 52 joints is 9×10^{-8}, the rounding of the single-precision samples. Fig. 1 shows them.

What is left out is stated plainly. The 2026 model trims the body, handle and spout against each other and adds chamfers at the joins; those surfaces come out of the generator as irregular meshes, not grids, and are not included, so the handle and spout still pass through the wall. The 1992 and 2006 versions, whose difference is in trimmed surfaces and interiors, are not included.

VII. Against the Later Teapots

Each later version fixed a named defect. We take them in turn and say what a node lattice does about each. Table III is the summary; the measurements follow.

TABLE III
Defects addressed by later versions, and by a CONV lattice

Defect of the 1976 patches	Fixed in	CONV lattice
Body not round	1992	Improved: 233 times rounder at the same node count
Kinked knob tip	mesh options	Improved: the kink is removed
No bottom	1987	Flat cap with an exact crease; Crow's round bottom from samples
Curvature jumps at patch joints	2026	Shape carried from samples; continuity itself not reproduced, see below
Handle and spout pierce the body	1992, 2026	Not addressed
No interior	1992–2026	The 2026 interior, from samples
Gap around the lid	1992	Not addressed
Mesh cracks, duplicate vertices, pole triangles	mesh options	Not applicable: no mesh is made
Texture layout	2026	One chart per part; patch coordinates also offered

A. Roundness

A cubic Bézier arc is not a circle, and the teapot's quarter arcs use a rounded handle length. At its widest ring the original body is out of round by $(r_{\max} - r_{\min})/\bar{r} = 4.1 \times 10^{-3}$. A node lattice needs no new patch type to do better: put the ring's nodes on the circle. A seamless periodic CONV ring through them is not a circle either, but its error falls quickly with the node count (Table IV). With the 16 nodes per turn of the exact lattice it is 233 times rounder than the original, and the body and lid stay within 4.3×10^{-5} of the exactly revolved profile. Because the profile factor is linear in the radius, the profile itself remains exact in this style.

TABLE IV
Out-of-roundness of the body at its widest ring

Surface	$(r_{\max} - r_{\min})/\bar{r}$
Bézier original, and CONV exact	4.1×10^{-3}
CONV round, 12 nodes per turn	9.6×10^{-5}
CONV round, 16 nodes per turn	1.7×10^{-5}
CONV round, 24 nodes per turn	1.6×10^{-6}
CONV round, 32 nodes per turn	2.8×10^{-7}

The same change repairs the knob tip. The original lid ends in a ring of radius 0.01 whose four quarter patches meet at a kink: the tangent around the ring turns by 18.9° a seventh of a node interval below the tip and by more nearer to it. Meshing options exist on the official page for such tips. On the round lattice the ring is a circle and the turn is zero.

B. Curvature continuity

Curvature continuity is the contribution of the 2026 model [6]: where two patches of the original meet with matching tangents but different curvature, a reflection bends. Fig. 2 repeats that test on our lattices. The 1976 lattice, being the 1976 surface, shows the bend at the joint of the upper and lower body. The 2026 lattice shows none at this scale.

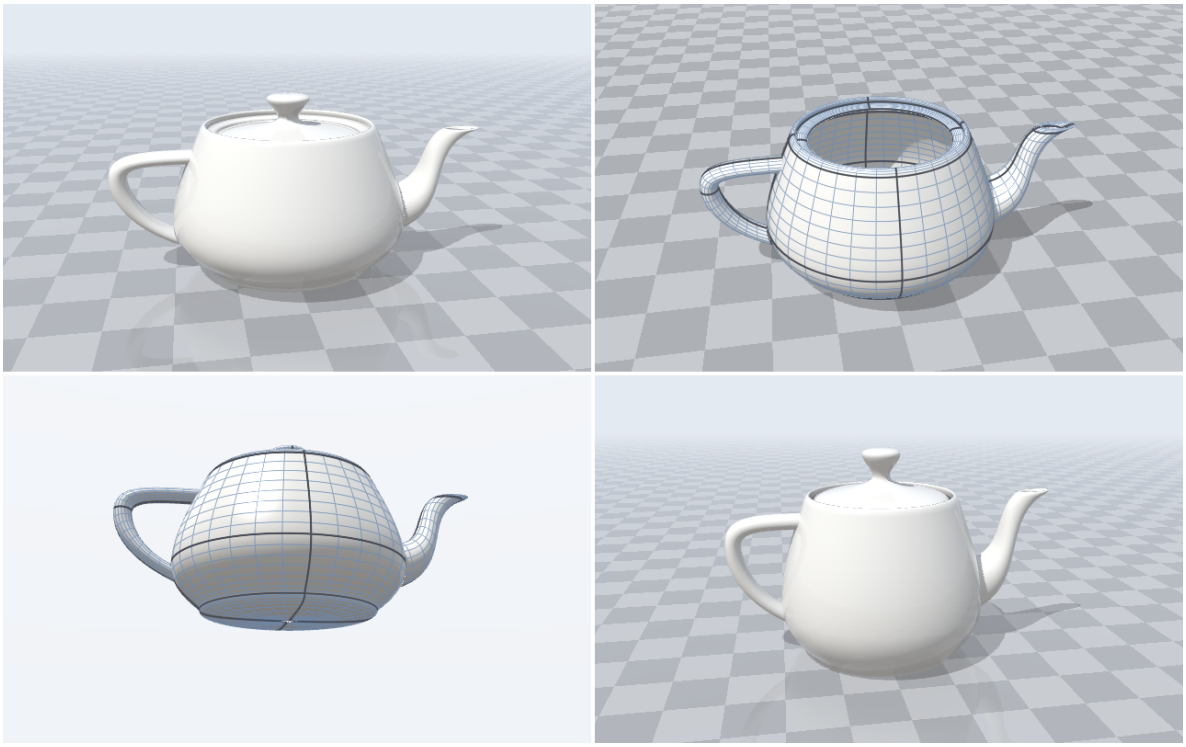


Fig. 1. Versions as node lattices. Top: the 2026 version from surface samples, glazed; the same with the lid removed, showing its interior and the node lines of each 9×9 lattice. Bottom: the 1987 version from below, with Crow's round bottom; the 1975 version in Newell's proportions.

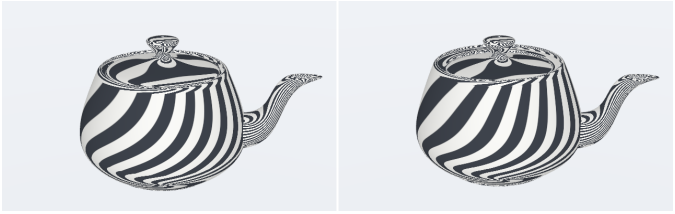


Fig. 2. Reflection lines. Left: the 1976 lattice; the stripes bend where the upper and lower body meet. Right: the 2026 lattice from samples.

The picture is kinder than the numbers, and the numbers are the result. At every joint between two exterior lattices we compared the two surfaces' normals and the curvature of the curve that crosses the joint (Table V). The sampled 2026 lattices keep the normal to 0.11° on average and 0.83° at worst. They do not keep the curvature: its relative jump across a joint averages 0.16, about what the 1976 patches have by design. The reason is the seam. Each lattice ends with one-sided stencils, whose second derivative is accurate to second order only, so two lattices that sample one G^2 surface still disagree about its curvature at their common edge. Our 2026 teapot therefore has the shape of Yuksel's, to the accuracy below, and approximately its normals, but it is not itself curvature continuous. Lattices that run across the smooth joints without a seam would be, where the limiter is inactive; they need samples whose spacing is compatible across the joint, which the generator's meshes do not promise. We leave that open.

TABLE V
Across the joints of exterior lattices, 9 by 9 nodes each

	Normal turn (deg)		Curvature jump	
Lattices	mean	max	max	mean rel.
1976, 44 joints	0.06	1.71	2.85	0.19
2026, 52 joints	0.11	0.83	49.4	0.16

The maximum for 1976 is the kink at the knob tip, and the large maximum for 2026 is on the narrow tubes, where curvature itself is large. Accuracy of the sampled shape. We have no finer truth for the 2026 surfaces than the samples, so we hold some out: lattices built from every other sample, 5×5 per patch, predict the omitted samples to 2.1×10^{-3} at worst and 6.6×10^{-4} rms on the exterior (3.7×10^{-3} and 8.0×10^{-4} on the interior), on a body 4.0 wide. The 9×9 lattices have half that spacing and fourth-order ends, so their error should be near a sixteenth of these figures; that is an expectation, not a measurement.

For the 1976 data alone the picture is as follows. Around the body and lid, the round style has no seams, and one-sided finite differences find no curvature jump across node lines (relative jump below 2×10^{-8}). Along the profile, the exact and round styles copy the original and therefore its curvature jumps: the worst is 2.85 in absolute curvature, at the joint of the rim and the upper body. They are not curvature continuous there, and Yuksel's construction [6] is.

One might hope to obtain continuity by removing the seams along the profile and leaving them only at true creases. We built that variant, with rows graded in arc

length so that spacing changes gradually. It is not a success at the node counts of interest. At four intervals per patch it deviates from the original by up to 4.0×10^{-2}, and it is not smooth either: where the tight curl of the rim falls between nodes, admission acts and leaves tangent turns of up to 13° at node rows, and the cap crease is rounded to 47° where it should be 90°. Doubling the nodes brings the deviation to about 2×10^{-2}. A CONV lattice is smooth where its data are resolved and it is honest about where they are not; it does not manufacture curvature continuity at a feature it cannot see. The seamed styles are the ones we recommend.

C. What is not addressed

The handle and spout still pass through the body in every version here; a lattice has no notion of trimming, and the trimmed and chamfered surfaces of the 1992 to 2026 models are not grids of samples. From outside the difference is confined to the joins, where the 2026 chamfers make the normal continuous and ours do not. In the 1975 to 1987 versions the lid still floats 0.1 above the rim.

D. No mesh

Several options of the official page concern tessellation: duplicate vertices along patch edges, triangles at the tips, matching normals across edges, the direction of triangulation. A node lattice has one node where patches meet, and Section VIII renders the surface itself. At a pole the solver's damped step and a normal taken a short distance along v are the whole treatment.

VIII. Rays by Transport

A camera ray meets $S(u, v)$ where two scalar equations vanish. The usual method searches for that point from the ray's side [9], [10]: a hierarchy of bounds, then Newton's method. Correspondence transport, introduced for a cloth experiment built on CONV, works from the surface's side. The parameters of a lattice of surface points are known; where those points land on the screen tells nearby pixels where their rays must meet the surface.

A. Method

1) Transport. Evaluate each part on a lattice of k samples per node interval ($k = 4$) and project the points.
2) Predict. Each projected lattice cell, split into two triangles and dilated by 0.6 pixel, gives every pixel centre it covers a candidate (u, v) by barycentric interpolation, with a lower bound on depth. A pixel keeps up to six candidates in depth order. Workers own disjoint bands of rows, so the lists need no locks and their order is deterministic.
3) Correct. In depth order, a pixel refines a candidate on the true surface by Newton's method with the dual-number derivatives, to 0.01 pixel. It stops at the first converged hit that is nearer than the next candidate's bound.

No ray searches anything. The triangles are predictors only; every hit, normal and texture coordinate comes from the surface.

Two refinements carry most of the gain. A prediction whose first residual is under 0.35 pixel takes one exact Newton step and is accepted without a second evaluation, provided the step stays inside one cell and is not long for its residual, which excludes creases and grazing rays; the error of such a step is of second order in the residual. And a solve that fails to halve its residual after two iterations is abandoned; without this, rays that just miss a silhouette cost eight evaluations each and dominate the frame.

B. Comparison

The baseline traces each ray independently through a bounding-volume hierarchy over the same lattice cells, intersects the same two triangles per cell, and corrects with the same solver and the same one-step rule. It is a strong baseline: it differs from transport only in how a ray finds its candidates. Table VI gives medians of 21 frames of the exact teapot on an Apple A18 Pro (six cores), Apple clang 21 at -O3 without contraction. The machine was shared during the measurements (load average near 6), so the absolute times are indicative and the ratios are the result.

TABLE VI
Frame time (ms), one sample per pixel

Frame	Threads	Transport	Per ray	Ratio
640×400	all	17.4	29.1	1.67
640×400	one	48.4	87.4	1.81
960×600	all	30.3	59.6	1.97
960×600	one	90.9	221.4	2.44

The mechanism is visible in the counters. At 960×600 transport starts 1.09 solves and makes 1.34 surface evaluations per surface pixel; the per-ray tracer starts 1.90 and makes 2.10, and visits 8.5 hierarchy nodes per ray besides. Transport pays 4.8 ms to predict and 4.7 ms to transport the lattice and build the shadow map. A repeat of the last row gave 183 ms for the per-ray tracer, so that ratio should be read as about two.

Agreement. For the 960×600 frame we re-evaluated every reported (u, v): the worst reprojection error is 0.20 pixel and the mean 5×10^{-4} pixel. Against a per-ray reference with a lattice four times finer and no one-step rule, transport agrees on 98,347 of 98,462 surface pixels (99.88%) and the per-ray tracer on 98,349. Nearly all the differences lie on silhouettes and crease edges, where a ray grazes two roots less than a pixel apart and neither method is certain to return the nearer. Neither proves that every intersection has been found.

C. Edges, floor, and gloss

Edges. One sample per pixel leaves stair-stepped silhouettes. A pixel whose neighbour shows another part,

another sheet or no surface receives four more samples on a rotated grid, which reuse the pixel's own predictions. About 2% of pixels qualify, but their rays are the expensive ones, and the frame costs about twice as much: 40.0 ms against 50.4 ms per ray at 640×400 on all threads.

Floor. The view reflected in a planar floor is the view from the eye mirrored in that plane, an identity used for planar mirrors in an earlier photonic transport experiment. Its rays are as coherent as the camera's, so the reflection is one more transport pass from the mirrored eye over the same lattice.

More surfaces. The 2026 version has 66 lattices and 5346 nodes, with interior sheets behind most exterior pixels. A 640×400 frame of it takes 38.6 ms by transport and 55.8 ms per ray without edge samples, and 61.1 ms and 80.3 ms with them.

Gloss. A ray reflected by the curved teapot is not coherent with its neighbours, and transport has nothing to offer it; those rays go through the per-ray tracer, one bounce. The emitter is a rectangle seen along the reflected ray, so a highlight has the shape of a window and not of a point light's lobe. With gloss and a polished floor a 640×400 frame takes 72.9 ms by transport and 96.7 ms per ray, edge samples included.

Wireframe and texture. A hit carries its own (u, v) and their screen gradients, so node lines and patch outlines are drawn in the shader with hidden lines removed for free, and they are curves on the surface, not a cage around it. Textures are small RGB lattices reconstructed by the same CONV evaluator, mapped per part or per patch as on the 1976 journal cover [2].

Fig. 3 shows four renderings and Fig. 4 the solver work per pixel.

IX. Motion

Since every node is a surface point, the collision shape of the teapot can be taken from the lattice with no separate model: the convex hull of a group of nodes is a convex piece of the teapot. For the 1975 to 1987 versions we use eleven hulls, one for the body, two for the lid, and four each along the handle and spout; for the 2026 version, one hull per group of exterior lattices. We hand them to the rigid-body engine of a rock-stacking game, whose wrench-transport solver resolves contact, friction and rolling resistance in one solve per frame. One model unit is taken as 7 cm. The engine returns a pose; the pose is applied to the rest nodes and the surface is recompiled through them, which takes 0.17 ms. Dropped from 42 cm at a tilt of 1.2 rad, the teapot lands on its spout, rocks back over its base and is asleep within 1.5 s (Fig. 5). The body is treated as solid porcelain, which is not how a teapot is made; a shell would need the interior that this model lacks.

X. Limits

The variation bound is factorwise and coordinatewise; it does not bound the curvature of the surface or forbid self-intersection. Exact reproduction is a statement about a rest pose whose admission is inactive; a deformed lattice is a CONV surface through the moved nodes, not a deformed Bézier surface. The surface is continuous within a part and only there. Transport and the per-ray tracer both accept a converged root without proving it the nearest, and both use lattice triangles as predictors, so a feature thinner than a lattice cell on screen can be missed. The hierarchy's boxes are padded by a fraction of their diagonal, which is a heuristic and not an enclosure. Timings are of one implementation on one shared machine. In a browser the engine runs on one thread in WebAssembly at about 80 ms for a plain 640×400 frame without edge samples.

XI. Conclusion

Put on a CONV lattice with seams, the Utah Teapot is the same surface it was in 1976, held by 560 points that lie on it. The lattice can be made round by moving nodes onto circles, traced without a mesh, textured in its own parameters, and dropped on a floor by handing its nodes to a physics engine. Later versions enter as samples, with no knowledge of how they were constructed: the 2026 teapot, interior included, is 66 small lattices. What the lattices do not carry across their seams is curvature continuity, and they do not carry trimming or chamfers at all. Those are the places where the later teapots, and Yuksel's in particular, did work that this representation does not replace. A seamless lattice over compatibly spaced samples of a G^2 surface is the experiment we would do next.

Attribution

The Utah Teapot was created by Martin Newell at the University of Utah and scaled by Jim Blinn; the 1987 bottom is Frank Crow's and the 2026 version is Cem Yuksel's. The models built here are Modified Utah Teapots in the sense of the terms on the official page [5]: the 1976 surface re-expressed, with an optional flat cap; and surfaces of the 1987 and 2026 versions rebuilt from points sampled with that page's mesh generator, without trimming or chamfers.

AI Generated Work Disclaimer

Substantial portions of this work were generated with the assistance of Anthropic Claude models.

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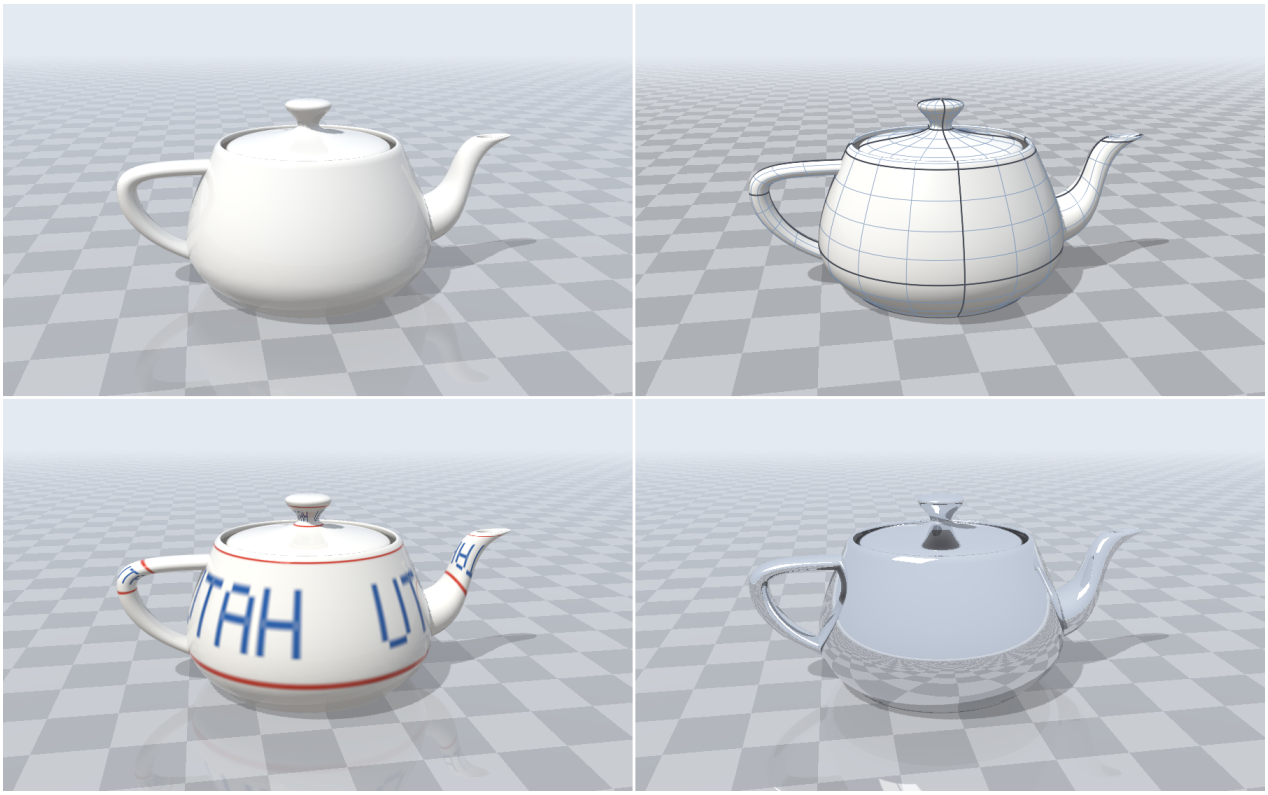


Fig. 3. The CONV teapot, exact style, 560 nodes, traced by transport. Top: porcelain with gloss on a polished floor; node lines (thin) and the original patch outlines (heavy) drawn from the hit parameters. Bottom: a low-resolution texture magnified by CONV under a traced glaze; chrome, reflecting the floor and its own handle and spout.

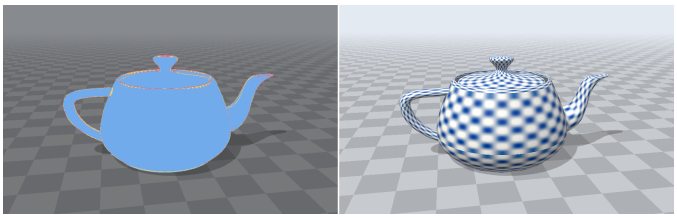


Fig. 4. Left: solver work. Blue pixels needed one solve, orange two, red more. Right: a checker mapped in patch coordinates.

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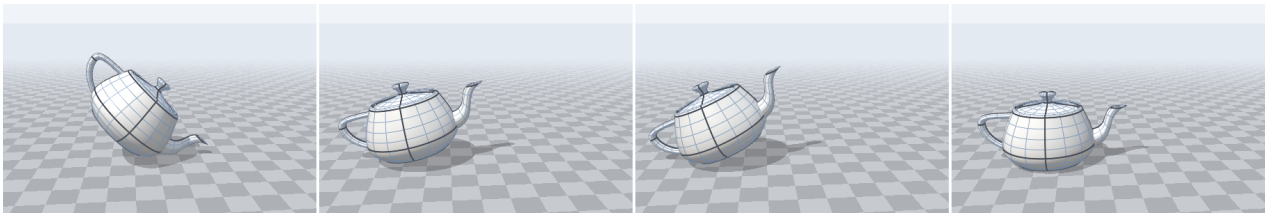


Fig. 5. A drop, at 0.30, 0.42, 0.60 and 1.60 s. The surface is rebuilt through the moved nodes at every frame.